I have *no* words: The development of truth-functional negation

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Background

- **Truth-functional negation**:
  - Negation inverts the **truth value** of a proposition.
  - Children understand the truth-functional nature of negation by at least **24-29 months**.[1, 2]
  - "Not" acquired at the same age.[1, 2]

- Children start producing negation around **15-17 months**.[3]
  - Single-word utterances like "no" [3, 4]
  - Used for rejection ("no veggies") and nonexistence ("no doggy?") [4]
  - Could reflect a more limited concept of negation [5]

- **"No"** can be used truth-functionally in **polar question** contexts
  - *Example:* asking "Is that a dog?" of a pig
  - Responding with "no" indicates "that is a dog" is false.

- Hummer et al. (1993): **20-23 months** old german-speaking children use "no" to answer polar questions.
  - Mothers read picture books with 4 polar questions to their child
  - No affirmative controls (asking "Is that a dog" of a dog)
  - No coding of gesture or cueing

- **Present study:**
  - Extend to English
  - Investigate children's ability to affirm [1]
  - Code for children's gestures and parent cueing [1]

Methods

- **Participants: 72 English-speaking children**
  - Young (16-19m, $\bar{x}$ = 18) → children start saying "no."
  - Middle (21-26m, $\bar{x}$ = 24) → some understanding of truth-functional neg
  - Old (28-32m, $\bar{x}$= 30) → strong understanding of truth-functional neg

- **Procedure:**
  - **Training:** Researcher reads a book and parents respond to questions.
  - **Testing:** Researcher leaves the room. Parents read two books that contain 16 polar questions (8 target-**no**, 8 target-**yes** trials) to their child.
    - Concrete words acquired early that were least likely to vary in referent

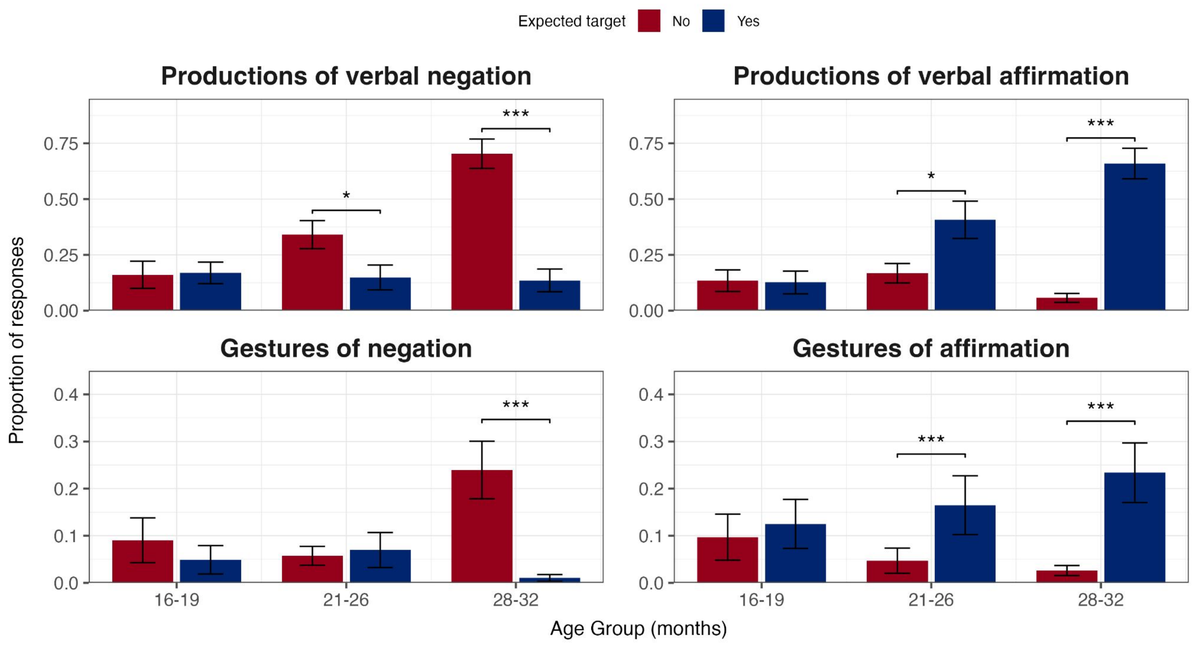




- **Coding:** Researcher **blind-codes** child & parent clips separately.
  - Child clips: verbal responses ("no", "yes"), gestures (head shakes, nods), labeling ("dog")
  - Parent cueing: Is this trial a target-yes or target-no trial?

- **Analysis:** Mixed effects regression models.

Results



- By **21-26 months**, children produced different verbal responses based on truth-conditions.
  - More negation on target-no trials (β = -2.33, SE = 1.05, p = .026)
  - More affirmation on target-yes trials (β = 1.59, SE = 0.73, p = .029)
- No response type differed by condition in 16-19 month olds (p>.10).

- Children responded with more correct **labels** on target-no than target-yes trials with age (β = 0.302, SE = 0.050, p<.001).
  - Middle (28% vs 12%, p<.001), Old (46% vs 10%, p<.001)

- Coding of **parent behavior** suggests no cueing.
  - Coder selected "neither target-yes or target-no" for 98% of the trials.

Discussion

- Children produce truth-functional "no" around **21-26 months old**.
  - Around the same time they begin understanding "not" [1, 2]
  - Might learn truth-functional use of "no" through co-occurrences with "not" (e.g., "no, that's not a dog")

- Affirmation and negation emerge simultaneously.
  - Is polar question comprehension developing?
  - Due to conceptual and linguistic linkage?

- Gestures did not occur earlier than verbal responses.
  - Consistent with gesture comprehension studies [1]

- 16-19 month olds do not produce negation and affirmation.
  - **Limitation:** only knew 71% of the words
  - Follow-up study using books with only words known by each child

- Cross-linguistic work could help disentangle decoding and conceptual explanations.[2]

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References: [1] Austin et al. (2014). Young children's understanding of denial. *Dev. Psychology*. [2] Feiman et al. (2017). What do you mean, no? Toddlers' comprehension of logical "no" and "not". *Lang. Learning and Dev.* [3] Frank et al. (2016). Wordbank: An open repository for developmental vocabulary data. *J. Child Lang.* [4] Bloom, L. (1970). *Language development: Form and function in emerging grammars.* [5] McDermott-Hinman & Feiman. (2025). The development of negation in language and thought. *Perspectives on negation: Views from across the language sciences*. [6] Hummer et al. (1993). On the origins of denial negation. *J. Child Lang.*